

PORT HEDLAND, Australia

WMO: 943120

Lat: 20.3725S

Lon: 118.6317E

Elev: 28

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 50.4	(c) 53.0	(d) 12.4	(e) 10.3	(f) 79.2	(g) 17.5	(h) 13.3	(i) 76.8	(j) 24.9	(k) 70.2	(l) 23.2	(m) 72.1	(n) 8.3	(o) 150	(p) 0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.4	105.7	70.4	102.6	71.2	100.4	71.9	83.2	91.9	82.1	90.2	81.3	89.3	14.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.4	86.7	80.3	157.6	86.5	79.1	151.4	85.8	47.3	92.3	46.1	90.3	45.2	89.7	98.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.9	(b) 22.6	(c) 20.7	(e) 46.0	(f) 111.9	(g) 2.7	(h) 3.2	(i) 44.0	(j) 114.2	(k) 42.4	(l) 116.0	(m) 40.9	(n) 117.8	(o) 38.9	(p) 120.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.8	87.4	87.2	87.2	83.4	75.7	69.9	68.6	70.7	75.2	81.5	84.4	86.7
	DBStd	8.11	3.60	3.69	3.70	3.61	4.22	4.46	3.91	3.46	3.91	5.01	4.30	3.76
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	23	0	0	0	0	0	9	12	2	0	0	0	0
	CDD50	10874	1159	1043	1153	1003	797	597	577	643	755	977	1031	1139
	CDD65	5421	694	623	688	553	332	156	123	180	305	512	581	674
	CDH74	66455	9414	8372	9216	6520	3071	1193	1139	1909	3279	6075	7286	8981
CDH80	34845	5164	4559	5145	3489	1327	283	246	626	1573	3401	4081	4951	
Wind	WSAvg	11.1	12.5	11.7	10.8	9.5	10.1	10.4	10.0	10.3	10.6	11.5	12.5	13.0
Precipitation	PrecAvg	10.9	2.3	4.0	1.6	0.6	0.9	0.9	0.4	0.2	0.0	0.0	0.0	0.2
	PrecMax	22.4	17.9	14.2	9.1	4.2	6.2	4.9	3.0	1.6	0.4	0.3	1.3	1.5
	PrecMin	2.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.2	3.5	3.5	2.3	1.0	1.5	1.3	0.7	0.4	0.1	0.1	0.2	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	107.4	107.3	106.4	102.6	95.6	89.2	88.2	91.7	99.2	106.8	108.0	108.0
		MCWB	71.8	74.0	71.1	69.6	64.9	65.6	61.6	62.4	63.7	65.8	67.2	68.9
	2%	DB	102.3	102.3	102.6	99.0	92.9	85.9	85.3	88.2	95.4	102.5	103.4	102.9
		MCWB	75.4	74.9	72.6	69.2	65.1	63.0	60.3	61.3	63.5	66.1	68.4	71.5
	5%	DB	98.4	98.7	100.0	96.8	90.0	83.2	82.6	86.0	91.8	98.7	100.0	99.0
		MCWB	78.5	76.6	73.9	69.4	65.0	61.9	59.7	60.1	63.4	66.8	68.9	74.2
	10%	DB	95.3	95.5	96.9	94.3	87.6	80.8	80.5	83.8	88.9	95.0	96.3	95.9
		MCWB	79.2	78.4	75.3	70.0	64.2	60.9	59.5	59.4	63.3	67.4	70.7	75.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	84.3	84.6	83.8	81.5	77.0	75.7	72.0	70.9	74.1	78.3	81.0	82.6
		MCDB	94.3	93.7	92.9	89.1	82.8	80.0	76.6	77.4	81.3	87.4	88.8	92.9
	2%	WB	82.8	83.1	82.1	79.3	74.5	71.7	69.0	68.5	71.3	76.5	79.1	81.0
		MCDB	91.5	91.7	89.3	85.9	80.8	76.8	75.5	75.9	81.3	85.0	88.2	90.6
	5%	WB	81.9	82.1	81.1	78.1	72.5	69.5	66.4	66.7	69.9	75.2	77.9	80.1
		MCDB	90.2	90.0	88.6	85.0	80.1	75.7	74.5	75.9	80.0	84.3	87.2	89.5
	10%	WB	81.0	81.1	80.3	76.8	70.6	67.1	64.2	64.8	68.5	73.7	76.6	79.2
		MCDB	89.3	88.8	87.8	84.4	78.5	74.6	72.8	75.0	78.8	83.4	86.3	88.4
Mean Daily Temperature Range		MDBR	16.4	16.7	19.1	22.5	23.1	22.5	24.8	27.4	28.0	26.5	23.1	18.5
	5% DB	MCDBR	22.4	22.5	24.5	26.1	26.4	24.2	27.2	30.0	32.6	31.9	29.6	24.9
		MCWBR	8.4	9.0	10.0	10.8	11.6	10.9	11.6	12.3	12.4	11.8	11.6	9.9
	5% WB	MCDBR	17.4	17.8	18.5	19.8	20.8	18.0	21.3	25.4	27.7	24.4	21.5	18.0
MCWBR		7.0	7.9	8.0	8.6	10.5	9.6	10.4	11.6	11.7	10.5	9.7	7.7	
Clear-Sky Solar Irradiance	taub		0.408	0.399	0.382	0.355	0.336	0.322	0.317	0.317	0.340	0.366	0.373	0.383
	taud		2.475	2.518	2.549	2.573	2.549	2.579	2.569	2.523	2.474	2.439	2.461	2.511
	Ebn at Noon		297	296	294	290	283	282	287	298	302	303	306	305
	Edh at Noon		38	36	33	30	29	27	28	31	35	38	38	36
All-Sky Solar Radiation	RadAvg		2305	2221	2006	1786	1497	1358	1466	1787	2105	2375	2531	2498
	RadStd		163	170	163	93	117	105	91	61	84	78	66	120

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.88	N/A	N/A	N/A	+0.86	+0.94	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.83	N/A	N/A	N/A	+0.86	+1.01	N/A	N/A	N/A

Nomenclature: See separate page